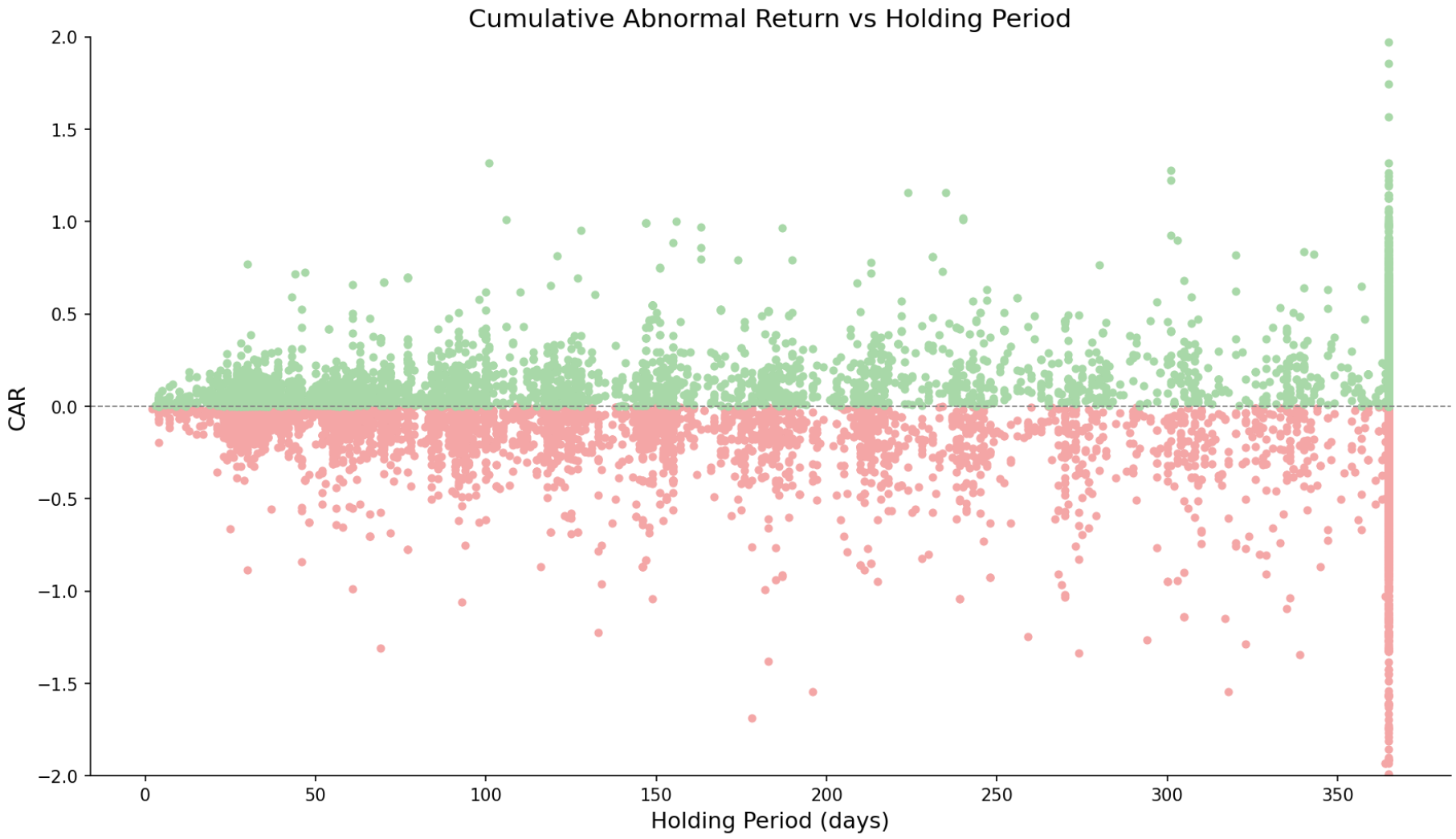
Congressional Stock Analysis

Do Congress Members Beat the Market?

Machine Learning on STOCK Act Disclosures · March 2026

Most Congressional Purchases Underperform the Market



15k
Total modeled trades

-0,5%
Median CAR

48.5%
Win rate

How We Measure Performance: Cumulative Abnormal Return (CAR)

Cumulative Abnormal Return (CAR)	
Positive CAR (+) Compensated MORE for risk than CAPM predicts	Negative CAR (-) Compensated LESS, paid for risk that did not materialize

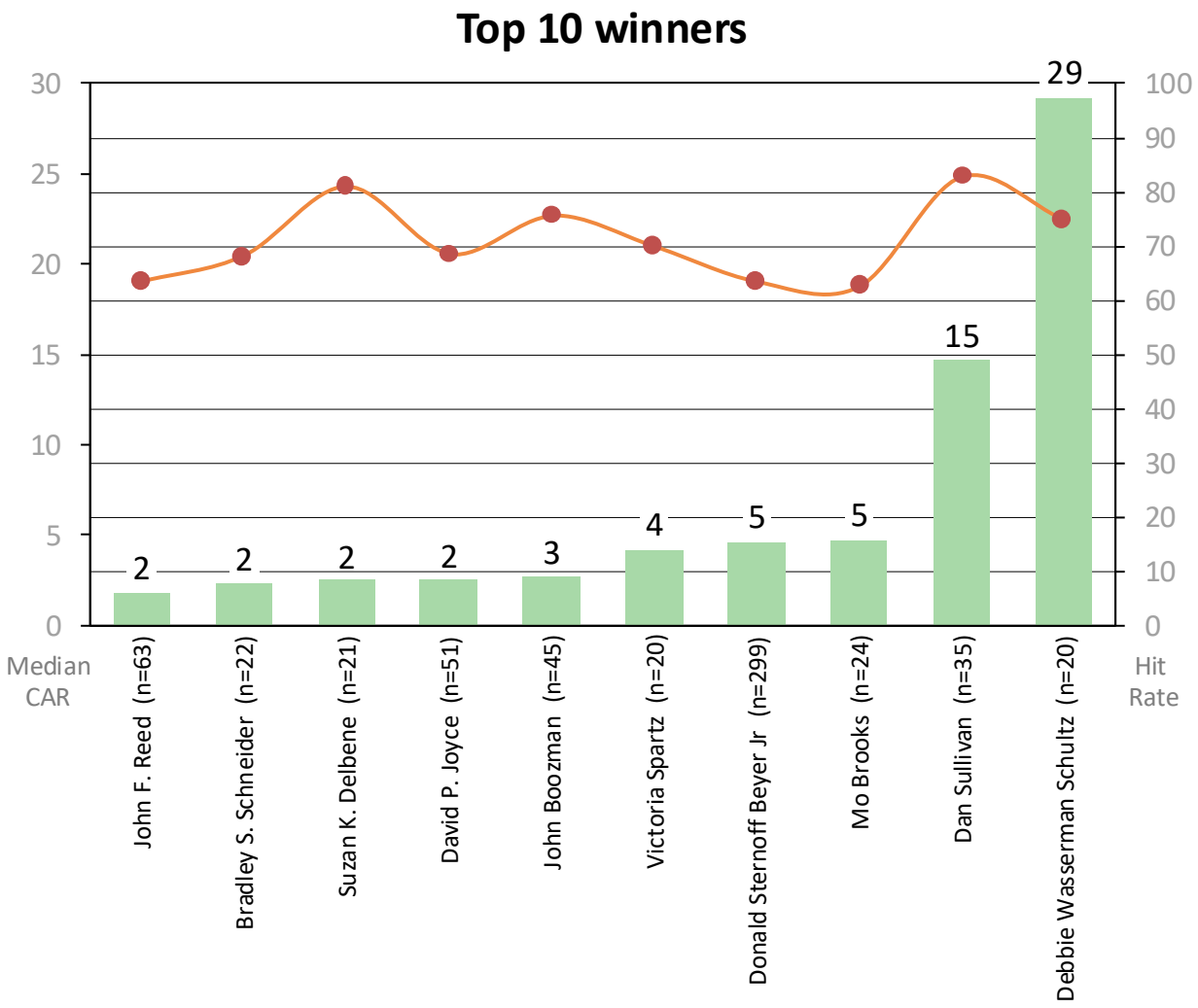
For each day, calculate the excess return relative to the stock's expected risk-adjusted return, then accumulate those daily excesses over the window.

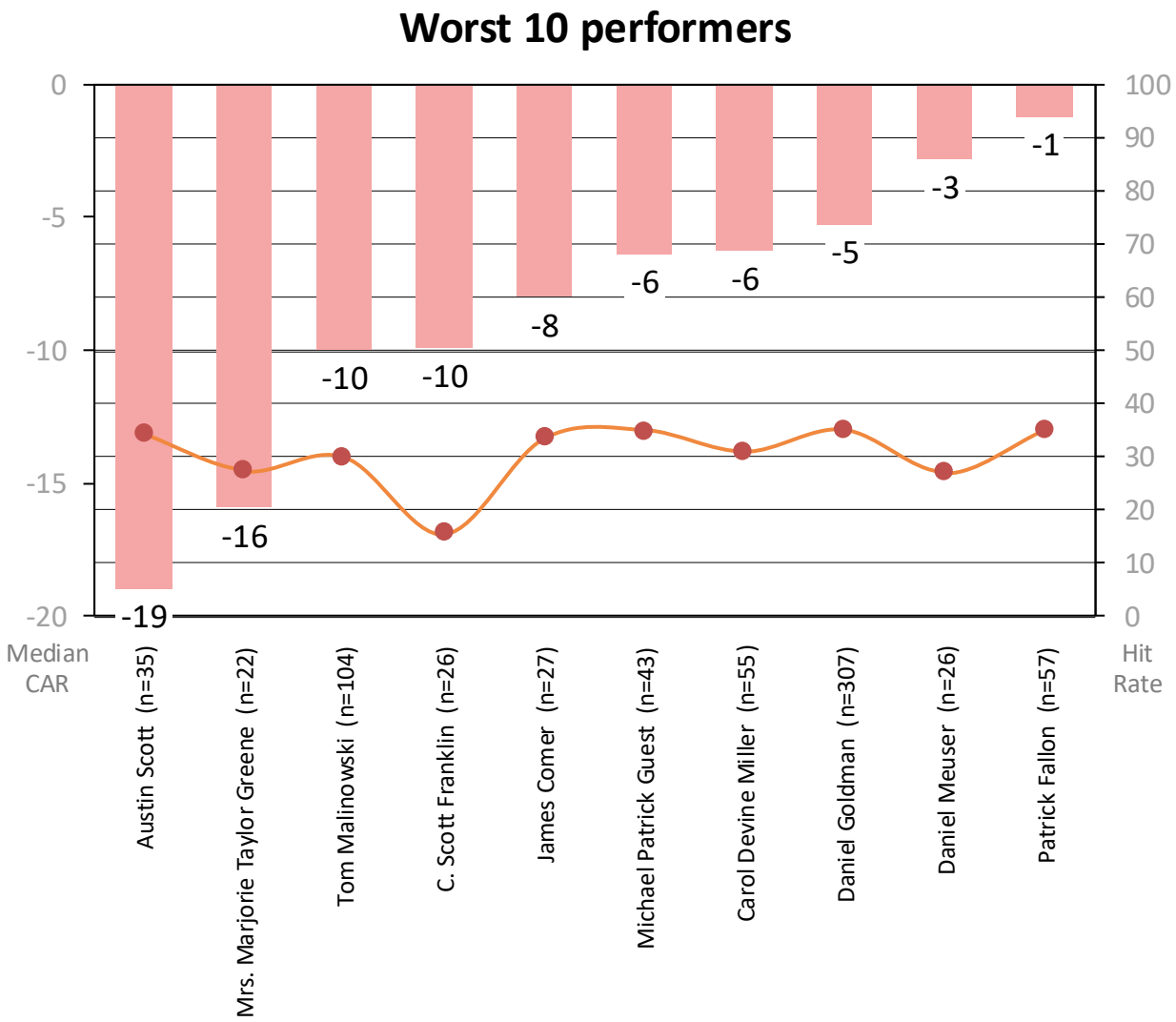
Excess return
actual log return –
CAPM-expected log return

CAPM
E[R] = R, + B x (Rm - R,)
Beta (B) estimated from 756 trading days of close prices (min 60 days).

Window
Measured buy-to-sell (if <12 months) or a fixed 12-month window.
The clock starts the day AFTER the public filing date

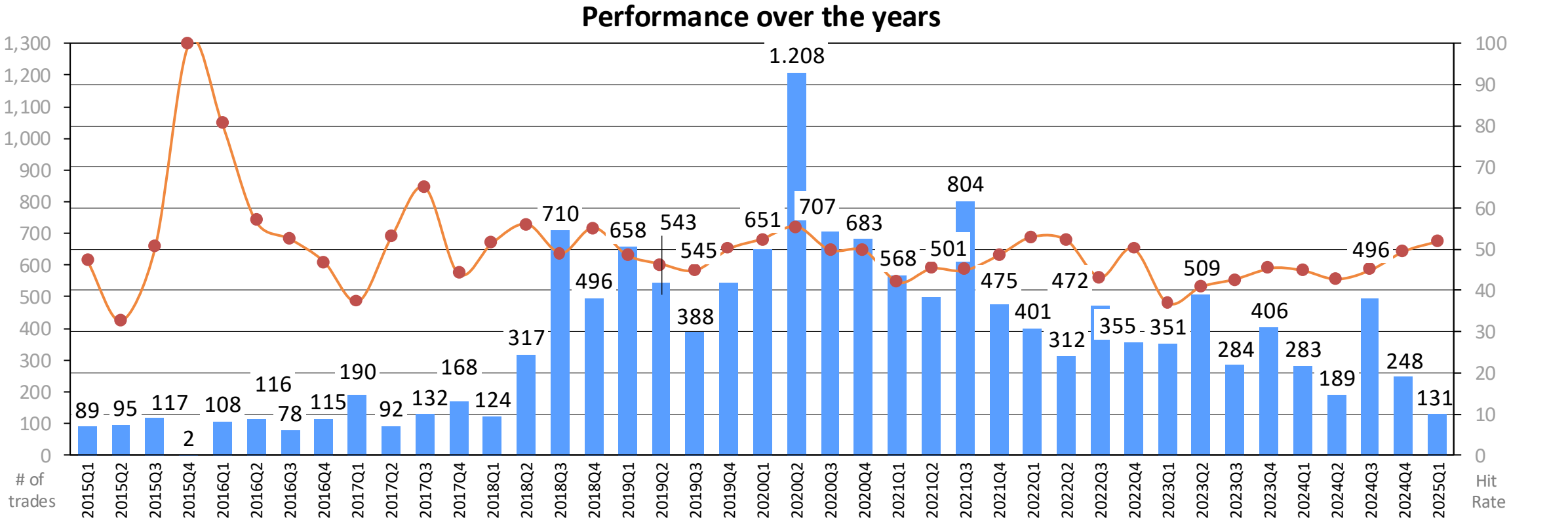
Who Consistently Beats — and Loses to — the Market?





- Based on 101 politicians with ≥ 20 disclosed purchases
- evaluated based on CAR performance, amount of $ traded was not considered

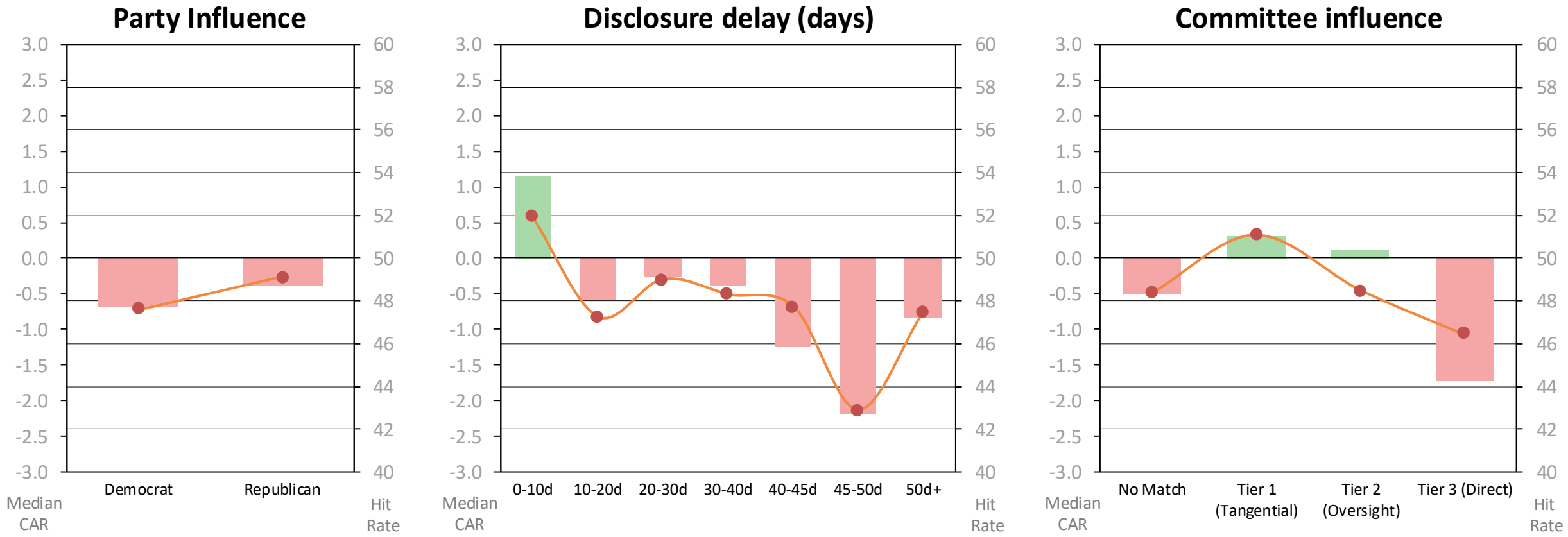
Performance Over Time: Macro Events Drive Positive Rates



- 2020 Q2–Q3: surge in positive rate — COVID recovery bounce lifted most purchases
- 2022: sharp drop aligned with rate-hike drawdown across the market
- Volume spikes suggest increased trading activity around election cycles
- Positive rate rarely exceeds 60% — even in the best macro environments

# of trades Hit Rate

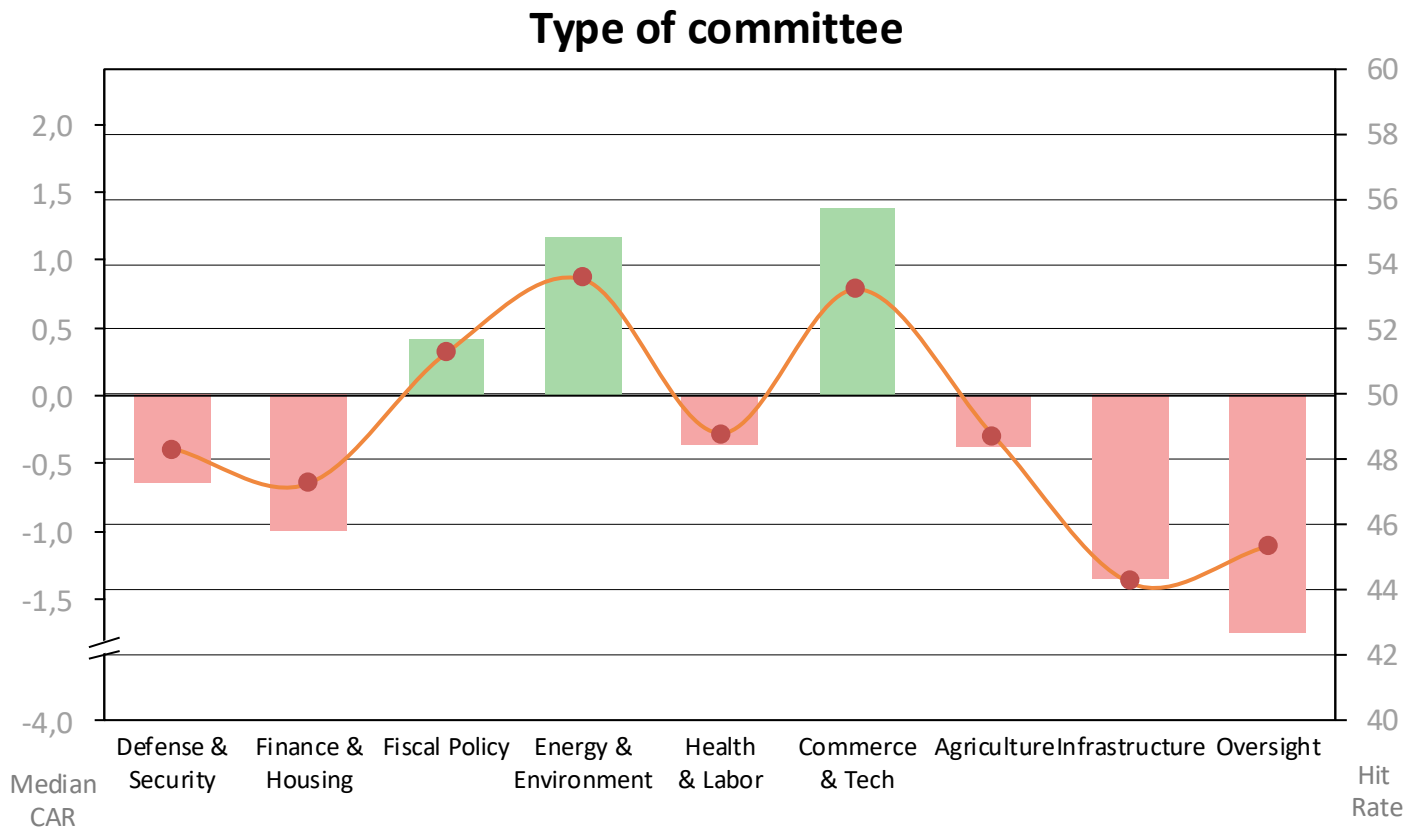
EDA: Median CAR Across Key Structural Variables

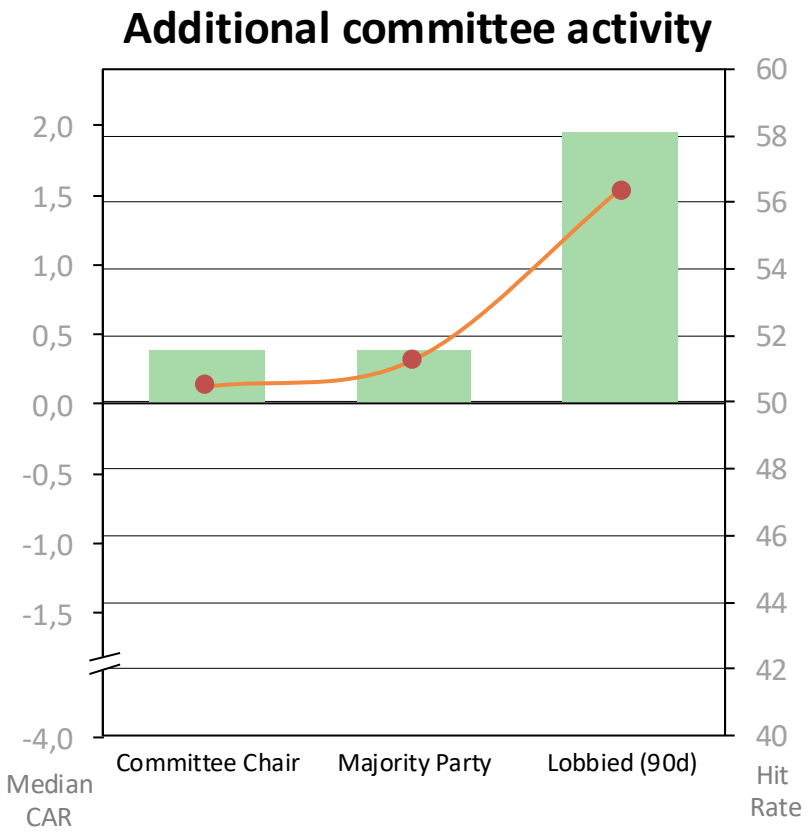


- Disclosure delay capped at 45 days by STOCK Act; 50d+ entries reflect late filers

Median CAR — Hit Rate

Committee Membership Shows Selective Advantage

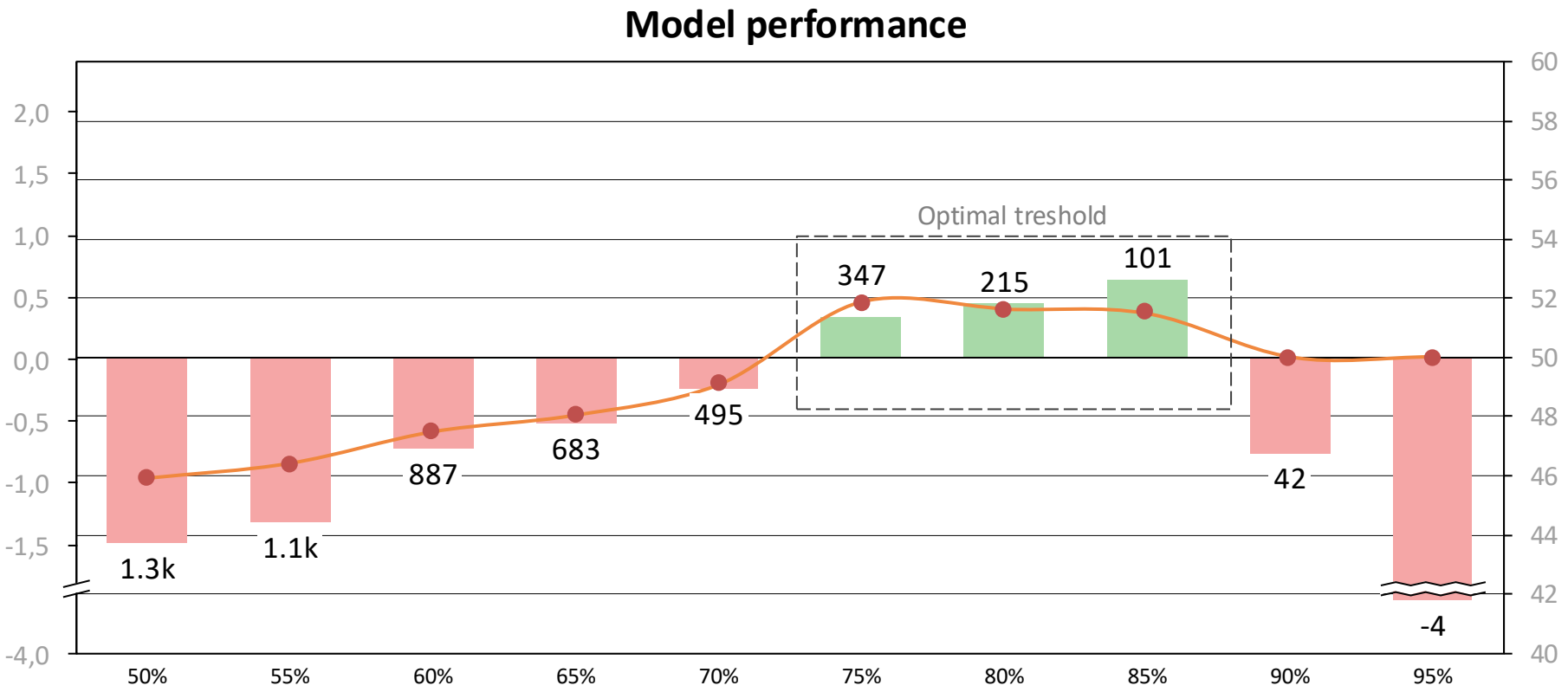




- Orange = flag=1 (on committee / in role / company lobbied) · Blue = flag=0
- All 9 committee categories shown — previously only 5 were displayed
- Lobbied (90d): company had active lobbying disclosure in 90 days before filing date

Median Car — Hit Rate

Can an AI Model do better?



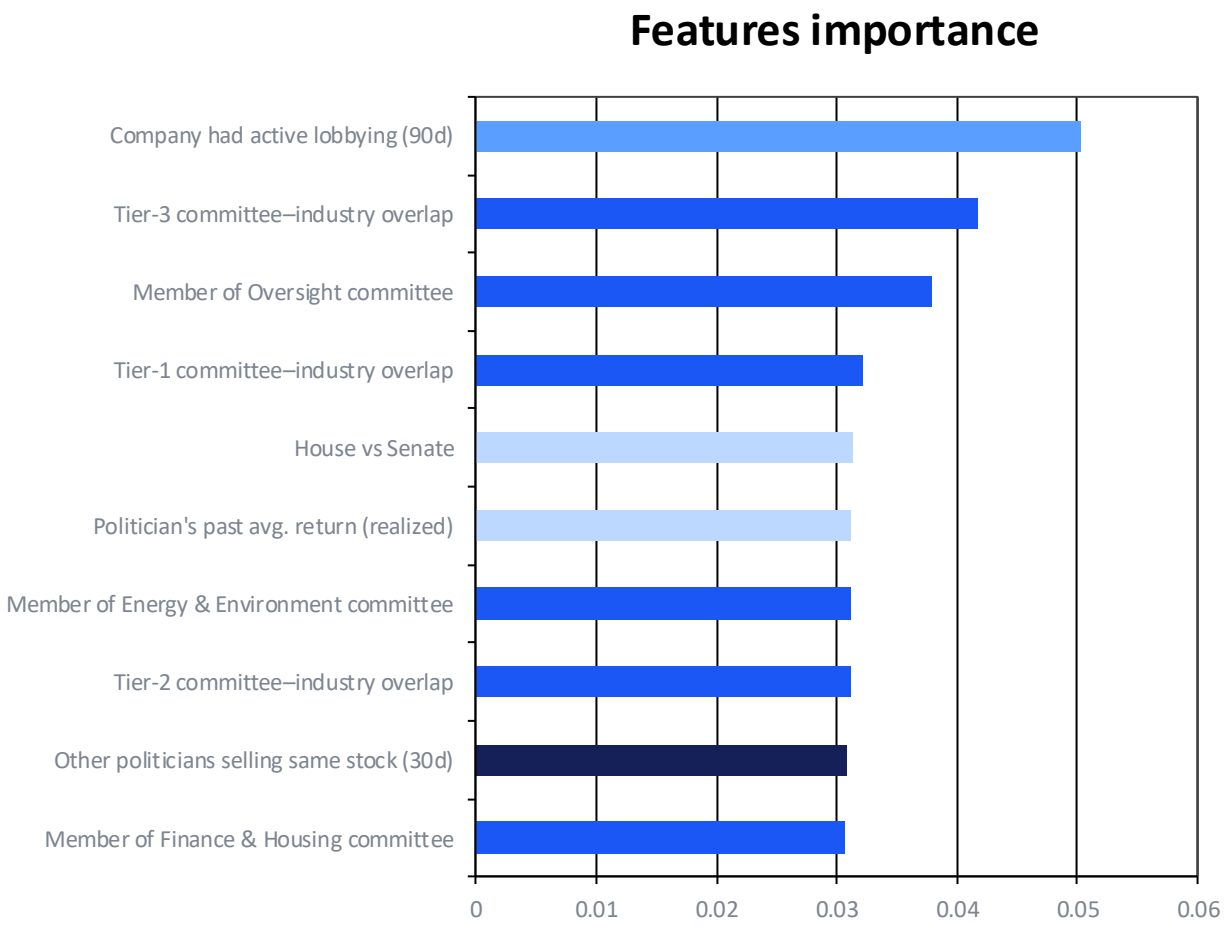
Optimal strategy:
The model's value lies not in predicting every trade, but in knowing which ones to ignore, filtering congressional purchases down to the highest-confidence signals flips a losing strategy into a marginally profitable one

2
Years of prediction data

+0,5%
Median CAR

52%
Win rate

The Model: What Drives the Predictions?



Politician characteristics

Past politician performance

Lobbying activity

Intersections between traded companies and active lobbying

Committee Jurisdiction

Committee–industry jurisdictional overlap

Stock Characteristics

Information related to the stock

Trade Signals

Specific trade information

Data sources

1: https://api.quiverquant.com/beta/bulk/congresstrading
2: Yahoo Finance
3: https://github.com/unitedstates/congress-legislators
4: https://api.quiverquant.com/beta/historical/lobbying

Conclusions & Caveats

Key Findings

- Most trades underperform: only 48.5% of purchases beat the market
- Past politician skill and lobbying activity are among the top model features
- Committee membership shows selective lift — most pronounced for Chair and Majority Party roles
- Tier-3 committee–industry match (direct jurisdiction) shows the strongest committee-related edge
- XGBoost reaches 53.0% precision at threshold 0.85 — fewer but higher-quality trades

Caveats

- No transaction costs or market impact modelled
- Test set is a single holdout period — results may not generalise across regimes
- Skill signals are backward-looking: past performance not guaranteed to continue
- CAR is measured from filing date — model already accounts for disclosure lag, but real-world execution is constrained to after public disclosure
- Dataset covers disclosed trades only — undisclosed activity not captured
- stocks that have been merged with other companies have been combined with the new owner or excluded

Potential Data Enrichment (Lobbying Focus)

- What more lobby data could add:
- 1. Lobbying spend amount & trend (currently: binary 90d presence only)
- 2. Issue specificity — which bills/topics overlap with politician's committee jurisdiction
- 3. Revolving door: lobbyists who previously worked for the same politician/committee
- 4. Proximity signal: days between lobbying start date and trade filed date

Thank you

Methodology: How Key Signals Were Constructed

1 — Ticker ↔ Committee Matching

- Committee membership from timestamped YAML snapshots (unitedstates/congress-legislators on GitHub)
- For each trade, the snapshot closest in time to the Filed date is used
- Ensures membership reflects the politician's actual committees at trade time
- Committee codes truncated to 4 characters to normalize across snapshot formats
- Result: binary committee-category flags per trade (9 categories total)

2 — 3-Tier Committee–Industry Match

- Manual mapping in config/commette_industry_map.yaml: committee → sector → industry
- Tier 3 (Strong): committee has DIRECT JURISDICTION — controls budgets, contracts, or legislation
- Tier 2 (Medium): committee has OVERSIGHT — sets general rules, taxes, or compliance standards
- Tier 1 (Tangential): committee has INDIRECT impact — general legislative influence only
- Result: Industry match 1/2/3 binary flags per trade (1 = politician's committee matches stock's industry at that tier)

3 — Lobbying Signal

- Source: QuiverQuant lobbying disclosures API, downloaded per traded ticker
- lobbied_any_90d = 1 if the company had any active lobbying disclosure in 90 days before trade's Filed date
- Currently binary — lobbying spend amount and issue specificity are future enrichment candidates

APPENDIX

Dataset Overview

39,334	19,096	15,117	268
Total Records	Purchases	Modelled	Politicians

Metric	Value
Date range	2012-09-13 – 2026-01-30
Transaction types	Purchase / Sale / Sale (Full) / Sale (Partial) / Exchange
Unique tickers (purchases)	2,209
Parties	D / R / I
Chambers	Representatives / Senate
Target variable	realized_car_hybrid > 0 (binary)
CAR windows available	1m / 3m / 6m / 9m / 12m + realized
Train / test split	80% / 20% chronological